Advanced Financial Theory

CFA Level II Style Questions & Solutions

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July 8, 2026

Section C Capital Structure and Strategic Finance

Q7: Capital Structure Conundrum

Context Aethelred Industries: Equity $3.5B, Debt $1.5B, R_E=14%, R_D=6%, T_c=21%

Requirements

1. Calculate R_A using MM Prop II with taxes.
2. Calculate WACC before and after tax.
3. Calculate tax shield value.
4. Explain trade-off and pecking order theories.
5. Recommend optimal capital structure.

Part (a): Unlevered Cost of Equity

$$R_E = R_A + \frac{D}{E}(R_A - R_D)(1 - T_c)$$

$$0.14 = R_A + 0.4286(R_A - 0.06)(0.79)$$

$$0.14 = R_A + 0.3386R_A - 0.0203$$

$$0.1603 = 1.3386R_A$$

$$\boxed{R_A = 11.97\%}$$

The unlevered cost of equity (11.97%) is lower than levered cost (14%) because leverage increases equity risk.

Part (b): WACC Calculations V = 3.5 + 1.5 = $5.0B

WACC Before Tax:

$$WACC_{BT} = \frac{3.5}{5.0}(0.14) + \frac{1.5}{5.0}(0.06) = 0.098 + 0.018 = \boxed{11.60\%}$$

WACC After Tax:

$$WACC_{AT} = \frac{3.5}{5.0}(0.14) + \frac{1.5}{5.0}(0.06)(0.79) = 0.098 + 0.0142 = \boxed{11.22\%}$$

After-tax WACC is lower because interest is tax-deductible.

Part (c): Tax Shield Value $V_L = E + D = 3.5 + 1.5 = \$5.0B$

$$V_U = V_L - T_c \times D = 5.0 - 0.21(1.5) = \boxed{\$4.685B}$$

$$PV(\text{Tax Shield}) = T_c \times D = 0.21 \times 1.5 = \boxed{\$315M}$$

MM Proposition I with Taxes: $V_L = V_U + T_c \times D$

$$5.0 = 4.685 + 0.315$$

The tax shield adds $315M to firm value.

Part (d): Trade-Off Theory

$$V_L = V_U + PV(\text{Tax Shield}) - PV(\text{Distress Costs}) - PV(\text{Agency Costs})$$

Benefits of Debt:

- Tax shield (interest is tax-deductible)
- Discipline on management
- Signals confidence

Costs of Debt:

- Financial distress (bankruptcy risk)
- Agency costs (debt-equity conflicts)
- Loss of flexibility (covenants)

Part (e): Pecking Order Theory Myers & Majluf (1984)

Financing Hierarchy:

1. Internal financing (retained earnings)
2. Debt (less information-sensitive)
3. Equity (adverse selection)

Key Assumption: Information asymmetry (managers know more).

Implications:

- No target debt ratio
- Financing driven by needs
- Profitable firms use less debt
- Equity issuance signals overvaluation

Part (f): Strategic Recommendation

D/E	R_D	R_E	WACC
0.43	6.0%	14.00%	11.22%
0.50	6.2%	14.35%	11.03%
0.60	6.5%	14.79%	**10.85%**
0.70	7.0%	15.23%	10.96%
0.80	7.5%	15.67%	11.12%

Optimal D/E = 0.60 (minimizes WACC)

Value Creation: Increases firm value by $1.19B

Q8: Dividend Policy Decision

Context Shares=100M, Price=$50, Earnings=$300M, Dividend=$2.00, Repurchase=$500M

Requirements

1. Compare dividend theories.
2. Calculate repurchase impact on EPS, price, value.
3. Analyze behavioral considerations.
4. Recommend optimal policy.

Part (a): Dividend Theories

1. **Dividend Irrelevance (MM)** Dividend policy does not affect firm value in perfect markets.
2. **Bird-in-the-Hand** Dividends are less risky than capital gains; investors prefer dividends.

$$P_0 = \frac{D_1}{R_E - g}$$

3. **Tax Preference** Investors prefer capital gains (taxed at 15% vs. dividends at 20%). Lower dividends → higher after-tax returns.

Q8 Solution: Repurchase Analysis

Part (b): Share Repurchase Analysis

$$\text{EPS} = \frac{300,000,000}{100,000,000} = \boxed{\$3.00}$$

$$\text{P/E} = \frac{50.00}{3.00} = \boxed{16.67x}$$

$$\text{Shares Repurchased} = \frac{500,000,000}{50.00} = \boxed{10,000,000}$$

$$\text{New EPS} = \frac{300,000,000}{90,000,000} = \boxed{\$3.33}$$

$$\text{New Price} = 3.33 \times 16.67 = \boxed{\$55.50}$$

Value Created: \$4,995M - \$5,000M $= \boxed{-\$5M}$

Part (c): Behavioral Considerations

Mental Accounting Investors view dividends as "income" and capital gains as "savings." Dividends are spent more readily.

Self-Control Dividends provide pre-commitment; retirees prefer dividends.

Signaling Dividend increases signal confidence (+2-3% reaction). Dividend decreases signal pessimism (-5-8% reaction).

Recommendation

1. **Optimal Payout:** Reduce from 66.7% to 50% ($1.50/share)
2. **Trade-off:**

Criteria	Dividend	Repurchase
Tax Efficiency	Lower	Higher
Flexibility	Lower	Higher
Signaling	Strong	Moderate

3. **Policy:** Increase dividend to $2.20/share + $200M buyback

Context Aethelred (MC \$15B, P/E 12.5x) acquiring Mercian (MC \$4.5B, P/E 22.5x)

FCF: Y1=150, Y2=180, Y3=216, Y4=259, Y5=311, TV=8,400 (WACC=10%)

Requirements

1. Calculate DCF value.
2. Analyze premium scenarios and synergies.
3. Evaluate behavioral biases.
4. Provide recommendation.

Part (a): DCF Valuation

$$PV(FCF) = \frac{150}{1.10} + \frac{180}{1.10^2} + \frac{216}{1.10^3} + \frac{259}{1.10^4} + \frac{311}{1.10^5}$$

$$= 136.36 + 148.76 + 162.28 + 176.34 + 193.06 = \boxed{816.80M}$$

$$PV(TV) = \frac{8,400}{1.10^5} = \frac{8,400}{1.6105} = \boxed{5,215.77M}$$

$$\text{Enterprise Value} = 816.80 + 5,215.77 = \boxed{\$6,032.57M}$$

$$\text{Equity Value} = 6,032.57 - 1,000 = \boxed{\$5,032.57M}$$

Part (b): Premium Scenarios

Premium	Price	Value ($M)	Premium
20%	$54.00	5,400	900
30%	$58.50	5,850	1,350
40%	$63.00	6,300	1,800

DCF value = $5,033M (11.8% premium). Premium ¿ 30% destroys value.

Part (c): Synergy Analysis **Revenue Synergies:** \$150M/year **Cost Synergies:** \$150M/year **Total:** \$300M/year

$$PV(\text{Synergies}) = \frac{300}{0.10} - 200 \text{ (integration)} = \boxed{\$2,800M}$$

Total Value: \$5,033 + \$2,800 = \$7,833M

Part (d): Behavioral Factors

CEO Hubris Overconfidence, empire-building

Winner's Curse Overpaying in bidding wars

Agency Problems Personal gain, firm size compensation

Recommendation

1. **Proceed?** Yes, but cautiously (strong strategic fit)
2. **Max Premium:** 30% ($58.50/share)
3. **Financing:** 50% cash, 50% stock
4. **Integration:**
    - Dedicated integration team
    - 100-day review
    - Clear KPIs

Section D Corporate Governance and Value Creation

Q10: Enterprise Risk Management

Context Aethelred implementing ERM framework after market volatility.

Requirements

1. Explain ERM principles.
2. Design ERM framework.
3. Calculate VaR and CVaR.
4. Discuss behavioral biases.
5. Explain ERM as counterweight.

Part (a): ERM Principles

1. **Integrated Risk View** Interconnected risks viewed holistically
2. **Portfolio Approach** Diversification reduces overall risk
3. **Strategic Alignment** Risk management supports strategy
4. **Continuous Monitoring** Dynamic and forward-looking

Part (b): ERM Framework

1. **Risk Identification** Strategic, Financial, Operational, Compliance
2. **Risk Assessment** Quantitative (P$\times$I), Qualitative (heat maps), Scenario analysis
3. **Mitigation** Avoid, Reduce, Transfer, Accept
4. **Monitoring** KRIs, Board committee, Regular reporting

Part (c): Risk Quantification

Risk	Prob	Loss ($M)	Expected Loss
Market	25%	200	50
Credit	15%	150	22.5
Operational	20%	100	20
Regulatory	10%	75	7.5
Strategic	30%	250	75
Total			**175**

$$\boxed{\text{VaR}_{95\%} = \$250M}$$

$$\boxed{\text{CVaR} = \$275M}$$

Part (d): Behavioral Biases

Overconfidence Underestimate extreme events

Anchoring Fixate on historical data

Confirmation Bias Seek confirming evidence

Availability Bias Overweight recent events

Part (e): ERM as Counterweight

1. **Independent Oversight** Separate risk function, board access
2. **Scenario Planning** Test tail risk scenarios
3. **Diverse Perspectives** External experts, dissenting views
4. **Decision Protocols** Structured approvals, documentation
5. **Post-Mortem** Review, learn, improve

Context Activist shareholder pressure on governance.

Current Compensation: Salary $3M, Bonus $5M (EPS-based), Equity 200k shares, Options 100k

Requirements

1. Explain governance elements.
2. Analyze agency conflicts.
3. Evaluate compensation.
4. Propose reforms.
5. Discuss activism.

Part (a): Governance Structures

Board Independent majority, independent chair, evaluations, diversity

Committees Audit, Compensation, Nominating, Risk

Compensation Align with long-term value, performance-based, clawbacks

Shareholder Rights Proxy access, majority voting, say-on-pay

Part (b): Agency Conflicts

1. **Shareholder vs Management** Compensation, perquisites, risk-taking
2. **Controlling vs Minority** Related-party transactions, voting rights
3. **Debt vs Equity** Asset substitution, underinvestment
4. **Free Cash Flow (Jensen, 1986)** Managers invest in negative NPV projects

Part (c): Compensation Reform

Component	Current	Proposed
Bonus	\$5M (EPS)	\$4M (Balanced)
Equity	200k shares	150k PSUs (ROIC)

Balanced Scorecard: Financial, Customer, Internal, Learning

Reforms: Clawbacks, 5-year vesting, no hedging, independent consultant

Part (d): Shareholder Activism

Strategies Board representation, capital return, strategic changes, governance reforms

Impact Increased accountability, transparency, value

Caution Potential short-term focus

Evidence Net positive impact; engage proactively

Context Pressure to integrate ESG factors into financial strategy.

Requirements

1. Compare shareholder primacy vs stakeholder theory.
2. Analyze ESG evidence.
3. Propose ESG framework.
4. Discuss future of ESG.

Part (a): Shareholder vs Stakeholder

Shareholder Primacy (Friedman, 1970) Only social responsibility is to increase profits

Stakeholder Theory (Freeman, 1984) Create value for all stakeholders (employees, customers, communities, environment)

Enlightened Shareholder Value Shareholder value maximized by considering stakeholders

Part (b): Empirical Evidence

Value Creation Positive correlation between ESG and performance

Cost of Capital Lower cost of equity and debt for high ESG firms

Risk Reduction Lower operational, reputational, regulatory risk

Investor Demand Growing demand for ESG investments

Part (c): ESG Framework

Environmental Carbon reduction (30% by 2030), renewable energy, sustainable supply chains

Social Employee welfare, community engagement, diversity

Governance Board oversight, ESG committee, sustainability reporting

Q12 Solution: Implementation

Part (d): Implementation Roadmap

Year 1: Assessment, baseline, materiality, stakeholder engagement

Year 2: Strategy, targets, action plans, pilot programs

Year 3: Full integration, ESG-linked compensation, reporting

Year 4+: Continuous improvement, benchmarking, feedback

Summary

Capital Structure MM foundation, trade-off theory, pecking order

Dividend Policy Irrelevance, bird-in-hand, tax preference, signaling

M&A DCF valuation, synergies, behavioral biases, integration

Governance ERM framework, agency conflicts, compensation, activism

ESG Stakeholder value, evidence, integration, implementation

Thank You

The modern finance professional must be both a quantitative analyst and a behavioral psychologist.

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